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Linear plasmid prevalence and linezolid resistance gene carriage in vancomycin-resistant *Enterococcus* in Canada from 2009-2024

Nicole Lerminiaux¹, Melissa McCracken¹, Jessica Bartoszko², Gurman Grewal¹, Sean Ahmed¹, Jennie Johnstone³, George R Golding¹, on behalf of the CNISP working group

¹National Microbiology Laboratory, Public Health Agency of Canada, Canada; ²Public Health Agency of Canada, Canada; ³Mount Sinai Hospital, Canada



INTRODUCTION

- Vancomycin-resistant *Enterococcus faecium* (VRE) is rising in hospitals in Canada¹, and resistance to last-resort antimicrobials including linezolid complicates treatment options for multidrug-resistant isolates
- Recent reports from around the globe indicate that both linezolid and vancomycin resistance genes can be mobilized by linear plasmids (named pELF) in *Enterococcus* species, and often on the same backbone²
- We aimed to investigate linear plasmid prevalence and linezolid resistance in VRE bloodstream infection isolates collected by the Canadian Nosocomial Infection Surveillance Program from 2009 to 2024**

METHODS

- Antimicrobial susceptibility testing (AST) was performed using the GPALL1F Gram-positive Sensititre panels (Thermo Fisher Scientific, USA)
- All isolates were sequenced with Illumina, and a subset was sequenced with Oxford Nanopore Technologies (linezolid-resistant by AST, n=21)
- Genomes were assembled with Shovill³ (Illumina) or Autocycler⁴ (ONT).
- pELF screening was done by mapping short-reads to the pELF1 reference plasmid ends using Bowtie2⁵ (--very-sensitive-local --no-unal), and at least five reads mapping to each end was required for a positive hit
- StarAMR⁶ for AMR & MLST, MOB-suite⁷ and pling⁸ for plasmid clustering, MBioSeq Ridom Typer for cgMLST, and panaroo⁹ for core gene analysis

RESULTS

- We obtained 2071 VRE isolates and 1.0 % (n=21) were linezolid resistant
- Almost half of isolates were predicted to carry linear pELF plasmids (45.4 %, 941/2071)
- The proportion of VRE with pELF plasmids has increased from 2018 (32.2 %, 59/183) and has stabilized around 72 % of isolates from 2021 to 2024 (2021: 68.5 % (115/168); 2022: 71.6 % (146/204); 2023: 72.8 % (166/228); 2024: 71.6 % (235/328)) (**Figure 1**)
- The increase in pELF prevalence was observed across the dominant sequence types (ST80, ST17, ST117)
- Most linezolid-resistant VRE (61.9 %, 13/21) carried a pELF linear plasmid; 46.2 % (6/13) encoded linezolid resistance genes (*optrA*, *cfr*(D), *poxA*), and 83.3 % (5/6) also had vancomycin resistance genes (*vanA*) (**Figure 2**)
- One pELF plasmid was 330.5 kb in length and was a co-integrate plasmid containing a ~237 kb insertion mediated by *IS1261E* (**Figure 3**)
- We clustered pELF linear plasmids from the global collection (n=154) with our plasmids (n=13) and found that most with linezolid resistance genes (93.3 %, 42/45) grouped in the same MOB-suite secondary cluster AK115
- We did a core gene alignment (n=42) and clustered using pling to identify nine subcommunities, and the mean number of structural changes (DCJ-Indel distance) between plasmids was 8.3 (**Figure 4**)

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CONCLUSIONS

- This study shows that linear plasmid prevalence is detectable and increasing across VRE in Canada**
- We provided further examples of co-carriage of vancomycin and linezolid resistance genes on mobile linear plasmids

Figure 1: Proportion of VRE isolates predicted to harbour pELF linear plasmids over time (top row, 2009 to 2024, n=2071) and by dominant sequence type (bottom row, 2018 to 2024)

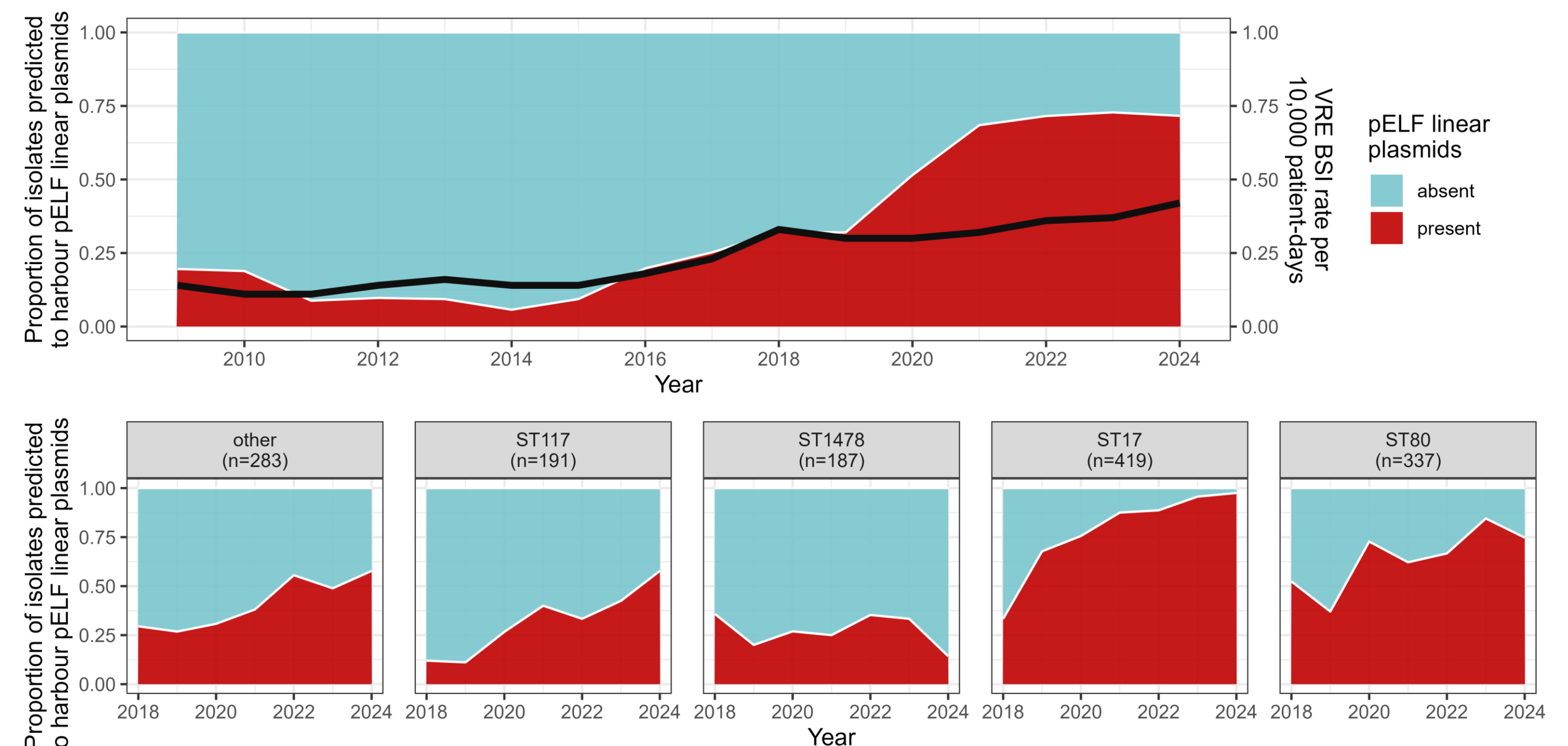


Figure 2: Features of pELF linear plasmids in linezolid-resistant VRE (n=13)

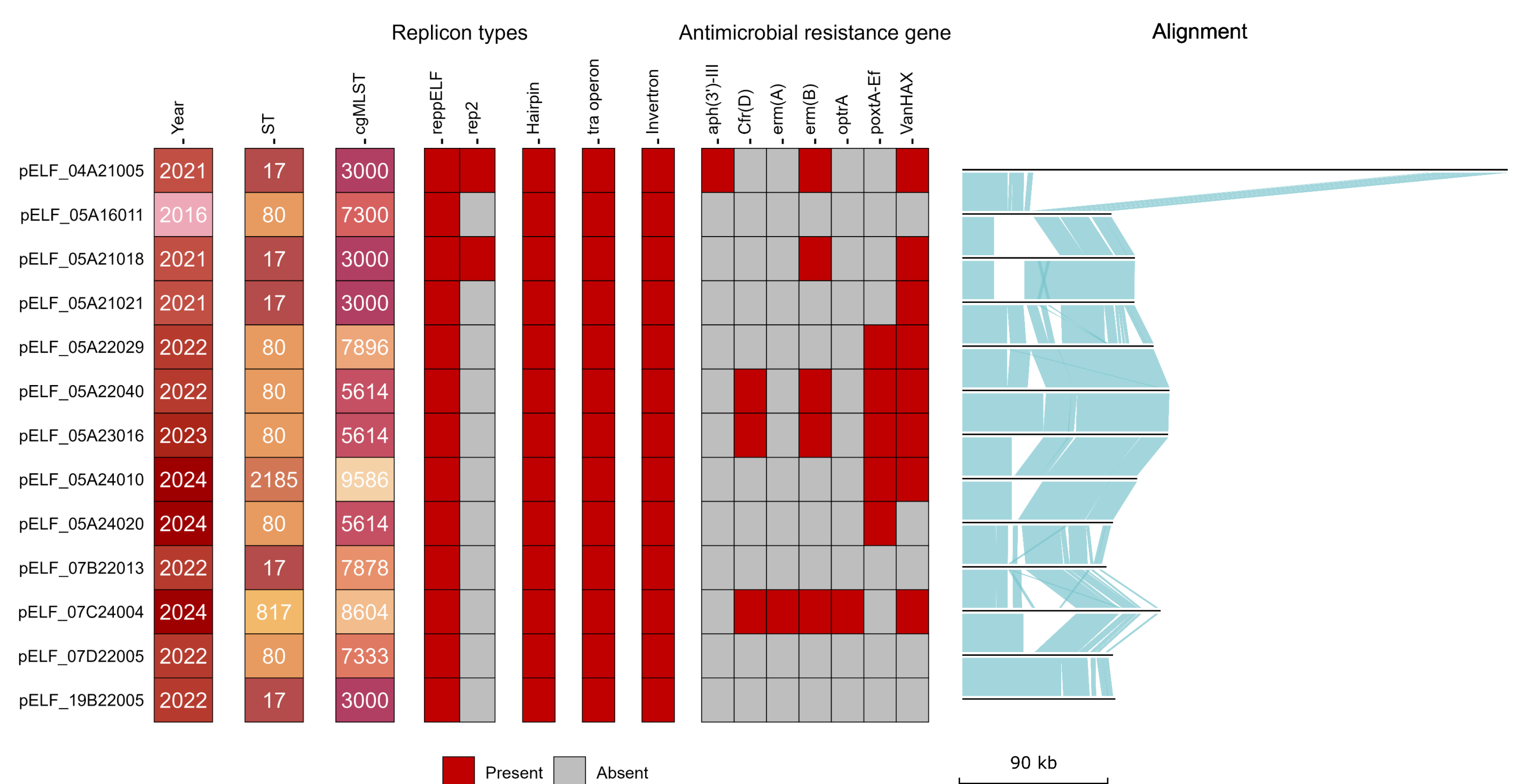


Figure 3: Map of co-integrate linear plasmid pELF_04A21005 (330.5 kb)

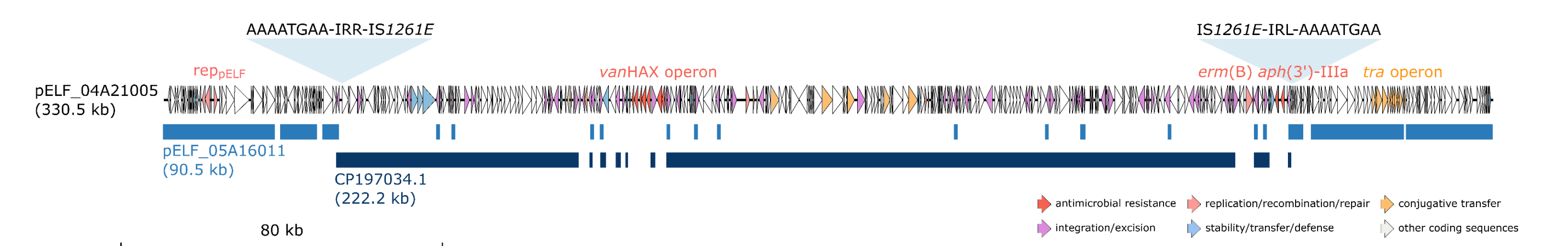
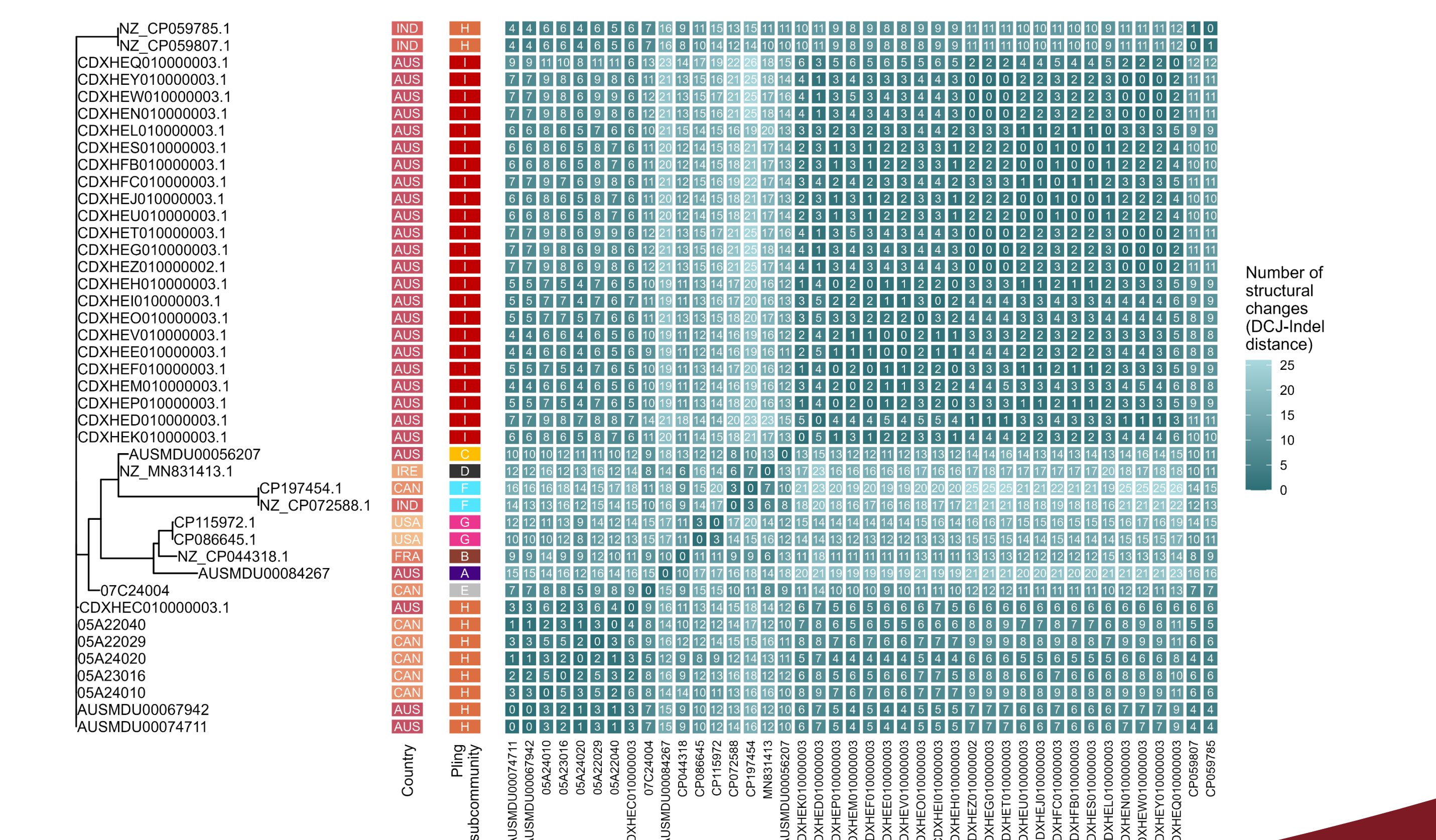


Figure 4: Core gene alignment and heatmap of pling DCJ-Indel distances of global & Canadian pELF plasmids encoding linezolid resistance genes (n=42)



References: ¹DOI:10.14745/ccdr.v51i67a04; ²DOI:10.1128/aac.01168-25; ³github.com/tseemann/shovill; ⁴DOI:10.1093/bioinformatics/btaf474; ⁵DOI:10.1038/nmeth.1923; ⁶DOI:10.3390/microorganisms10020292; ⁷DOI: 10.1099/mgen.0.000206; ⁸DOI:10.1099/mgen.0.001300; ⁹DOI: 10.1186/s13059-020-02090-4